WPL Performance Analytics Dashboard (2023–2026)

Project Documentation | Power BI Desktop 2.159.883.0

1. Project Overview

This project analyses Women's Premier League (WPL) match data across the 2023, 2024, 2025 and 2026 seasons. The report is presented as a three-page interactive Power BI dashboard covering competition overview, individual player performance, and team performance.

The project uses Python for data preparation and Power BI for data modelling, DAX calculations and visual analytics.

2. Business Objectives

Summarise WPL competition activity across four seasons.

Compare team batting output, bowling performance and match wins.

Identify leading batters and bowlers using player-level statistics.

Provide interactive views that help users explore team, season and player performance.

3. Dataset and Scope

Source: Cricsheet ball-by-ball cricket data.

Coverage: 88 matches across four seasons (22 matches per season), 2023–2026.

Teams represented: Mumbai Indians (MI), Delhi Capitals (DC), Royal Challengers Bengaluru (RCB), Gujarat Giants (GG) and UP Warriorz (UPW).

Data was converted from Cricsheet YAML files into tabular CSV datasets for analysis. The project also uses prepared match, delivery, team and player summary tables.

4. Tools and Technologies

Python — data extraction, transformation and preparation.

Pandas — tabular data processing and summary preparation.

Power BI Desktop 2.159.883.0 — data modelling, DAX measures and dashboard development.

DAX — calculated measures and dynamic KPI logic.

Cricsheet — ball-by-ball match data source.

5. Data Preparation

Cricsheet YAML match files were converted into structured tabular data.

Match-level and delivery-level data were organised into separate datasets.

Team and player names were standardised for consistent grouping, including consolidating Royal Challengers Bangalore under Royal Challengers Bengaluru (RCB).

Prepared summary tables were created for team performance, season batting, season bowling, batting player statistics and bowling player statistics.

Data checks were performed against expected match counts and aggregate statistics before dashboard validation.

6. Data Model and Main Tables

matches_cleaned — match-level information, including season and winner.

team_performance — overall team totals such as runs, wickets, runs conceded, runs per innings and economy.

season_batting — season-wise team batting summaries.

season_team_bowling — season-wise team bowling summaries.

batting_summary — player-level batting statistics.

bowling_summary — player-level bowling statistics.

best_batter_each_season and best_bowler_each_season — seasonal leader summaries.

DimSeason — season dimension used by the overview page slicer.

BattingPlayerImages and BowlingPlayerImages — player selection and image lookup tables.

7. Dashboard Pages and Features

- Page 1 — WPL Overview: competition KPIs, season selection, leading batter and bowler highlights, team runs, team wins and bowling economy insights.
- Page 2 — Player Performance: batter and bowler selection, dynamic player images, batting KPIs (runs, average, strike rate, fours and sixes), bowling KPIs (wickets, economy, average and overs), and top-five player charts.
- Page 3 — Team Performance: overall team KPIs and comparisons, team batting and bowling comparison, bowling economy, bowling average, season-wise team runs, season-wise bowling economy and season-wise wins.

8. Key Data Highlights

Matches analysed: 88.

Seasons analysed: 4 (2023–2026).

Teams analysed: 5.

Overall team runs in the prepared team summary: 27,378.

Bowler-credited wickets in the prepared team summary: 1,000.

Team wins recorded: 87, with one tied match having no winner.

Leading batter across the supplied dataset: Nat Sciver-Brunt — 1,348 runs.

Leading bowler across the supplied dataset: Amelia Kerr — 54 wickets.

9. Validation Notes

Dashboard KPI values and chart totals were cross-checked against the prepared datasets.

The project distinguishes bowler-credited wickets (1,000 in the team summary) from all recorded dismissals (1,079); these are not interchangeable.

Team totals and season-level summaries are presented at their respective aggregation levels. The season slicer is intentionally used on the Overview page; player and team pages retain their own page-specific interactions rather than forcing a global slicer across every visual.

10. Conclusion and Future Enhancements

The dashboard provides a multi-season view of WPL team and player performance and demonstrates data preparation, analytical modelling, DAX and Power BI visualisation skills.

Possible future enhancements include adding new seasons when data becomes available, expanding player comparisons, adding drill-through pages, and publishing a refreshed version with updated match data.

11. Repository Contents (Suggested)

- data/ — cleaned or sample datasets, subject to source licensing and file-size considerations.

- notebooks/ — Python data extraction and preparation notebooks.
- dashboard/ — Power BI .pbix file, if sharing is permitted.
- screenshots/ — overview, player performance and team performance page images.
- README.md — project summary and setup notes.